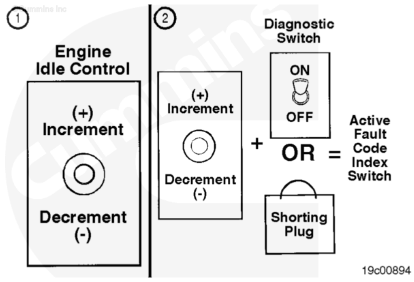
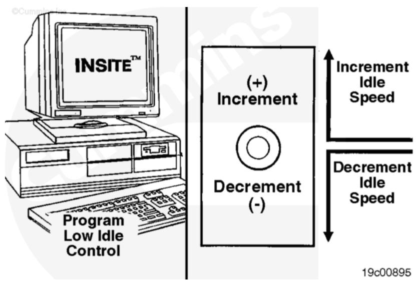
Trainings

General Information

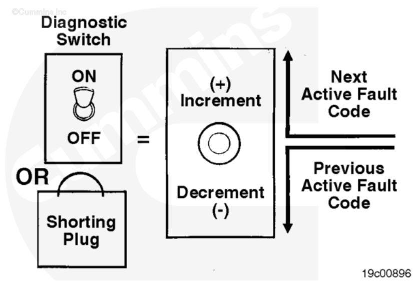
The idle adjustment feature is a part of the cruise control set/resume multi-functionality switch. Moving the switch to the set position increases the low idle speed and moving the switch to the resume position decreases the low idle speed.



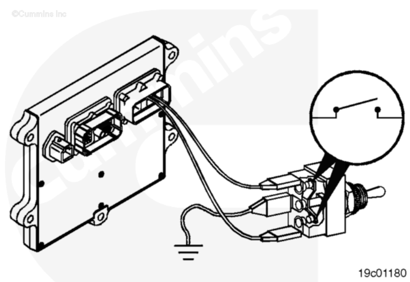
Depending on how the switch is configured, moving the switch in one direction will increase the low idle speed.



Push the diagnostic switch to the ON position or install the shorting plug. After the first active fault code has flashed out, push the idle adjust switch positive (+) up to advance to the next active fault code. Push the switch again until all of the active fault codes have been recorded.



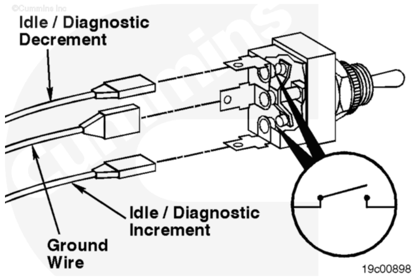
The idle adjust switch circuit consists of the idle/diagnostics increment signal, the idle/diagnostics decrement signal, the return wire, and the two-position switch located in the vehicle.



Resistance Check

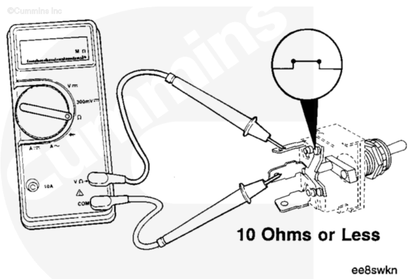
If an electronic service tool is available, monitor the idle adjust switch for proper operation. If **not,** follow the troubleshooting procedures in this section.

Remove the three electrical connectors from the switch. Label the wires with the switch location and the circuit name.



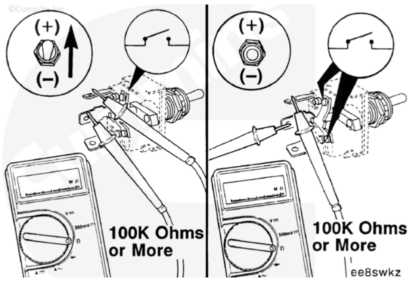
Touch one probe of the multimeter to the center terminal of the switch.

Touch the other probe to the cruise control/PTO resume/accelerate switch signal terminal of the switch.



Hold the idle adjust switch in the positive (+) increment position. The multimeter **must** show an open circuit (100k ohms or more) when the switch is held in the positive (+) increment position and after it is released. If the circuit is **not** open, the switch has failed.

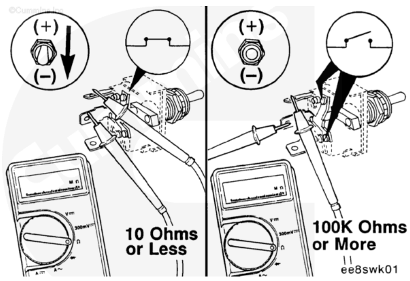
Refer to the OEM troubleshooting and repair manual for the replacement procedures.



Hold the switch in the negative (-) decrement position. The multimeter **must** show a closed circuit (10 ohms or less) when the switch is held in the negative (-) decrement position.

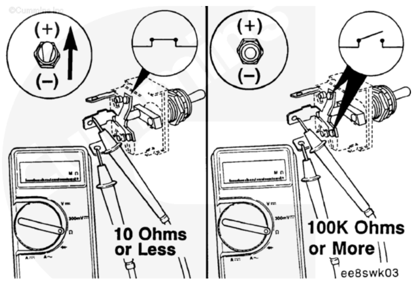
When the switch is released, it **must** show an open circuit (100k ohms or more). If the multimeter does **not** show the correct values, the switch has failed.

Refer to the OEM troubleshooting and repair manual for the replacement procedures.



Move the electrical lead from the cruise control/PTO resume/accelerate switch signal terminal to the cruise control/PTO set/coast switch signal terminal.

Hold the idle adjust switch in the positive (+) increment position. The multimeter **must** show a closed circuit (10 ohms or less) while the switch is held in the positive (+) increment

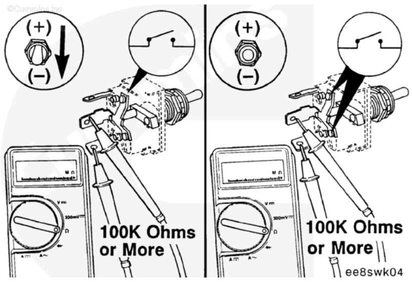


position.

When the switch is released, the multimeter **must** show an open circuit (100k ohms or more). If the multimeter does **not** show the correct values, the switch has failed.

Refer to the OEM troubleshooting and repair manual for the replacement procedures.

Move the idle adjust switch to the negative (-) decrement position. The multimeter **must** show an open circuit (100k ohms or more) when the switch is held in the negative (-) decrement position and when it is released. If the circuit is **not** open, the switch has failed.

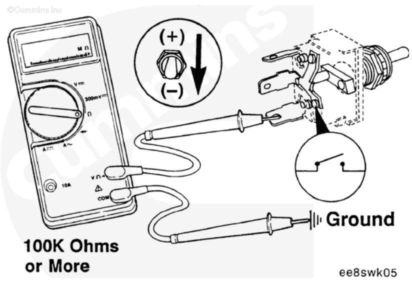


Refer to the OEM troubleshooting and repair manual for the replacement procedures.

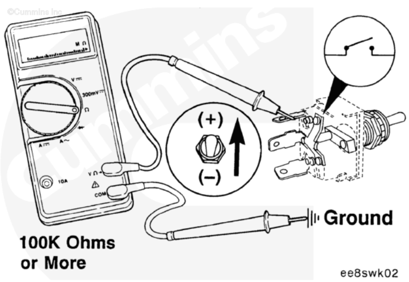
If the resistance value is correct, the switch **must** still be checked for a short circuit to ground.

Check for Short Circuit to Ground

Touch one multimeter probe to the cruise control PTO set/coast switch signal terminal of the switch and touch the other multimeter probe to chassis ground. Move the idle adjust switch to the negative (-) decrement position then to the positive (+) increment position. Measure the resistance. The multimeter **must** show an open circuit (100k ohms or more) when the switch is in all positions. If the circuit is **not** open, the switch has failed. Refer to the OEM troubleshooting and repair manual for the replacement procedures.



Check for a short circuit to ground. Remove the multimeter probe from the cruise control/PTO set/coast switch signal terminal and touch it to the cruise control/PTO resume/accelerate switch signal terminal of the switch. Keep the other multimeter touching chassis ground. Move the switch to the positive (+) increment position then to the negative (-) decrement position. Measure the resistance. The multimeter **must** show an open circuit (100k ohms or more) when the switch is in all positions. If the circuit is **not** open, the



switch has failed. Refer to the OEM troubleshooting and repair manual for the replacement procedures. If the switch passes all of the previous checks, the switch circuit **must** be checked.

Last Modified: 29-Jun-2015